

Unit 6. Orthogonality and Projection

1 Orthonormal bases and QR

1.1 Orthonormal sets

Definition 6.1 (Orthonormal set). A finite list $\mathbf{q}_1, \dots, \mathbf{q}_k \in \mathbb{R}^n$ is **orthonormal** if

$$\mathbf{q}_i' \mathbf{q}_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

For $\mathbf{Q} = (\mathbf{q}_1, \dots, \mathbf{q}_k)$, this condition is $\mathbf{Q}'\mathbf{Q} = \mathbf{I}_k$. The empty list is orthonormal.

Origin: H §6.2a

Definition 6.2 (Frobenius inner product and norm). For $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}$, define

$$\langle \mathbf{A}, \mathbf{B} \rangle_F = \text{tr}(\mathbf{A}'\mathbf{B}) = \sum_{i=1}^m \sum_{j=1}^n a_{ij}b_{ij}, \quad \|\mathbf{A}\|_F = \langle \mathbf{A}, \mathbf{A} \rangle_F^{1/2}.$$

Thus

$$\|\mathbf{A}\|_F^2 = \text{tr}(\mathbf{A}'\mathbf{A}) = \sum_{i=1}^m \sum_{j=1}^n a_{ij}^2.$$

Origin: Course formulation from H §6.1b, Eqs. (6.1.5)–(6.1.6)

Theorem 6.3 (Orthogonal nonzero vectors are independent). If $\mathbf{v}_1, \dots, \mathbf{v}_k$ are nonzero and $\mathbf{v}_i' \mathbf{v}_j = 0$ for $i \neq j$, then they are linearly independent.

Origin: H Lemma 6.2.1

Proof. If $\sum_{j=1}^k a_j \mathbf{v}_j = \mathbf{0}$, then, for each i ,

$$0 = \mathbf{v}_i' \sum_{j=1}^k a_j \mathbf{v}_j = a_i \mathbf{v}_i' \mathbf{v}_i = a_i \|\mathbf{v}_i\|^2.$$

Since $\|\mathbf{v}_i\|^2 > 0$, $a_i = 0$ for every i . □

Theorem 6.4 (Gram–Schmidt orthonormalisation). Let $k \geq 1$ and let $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$ be linearly independent. Define

$$\mathbf{u}_1 = \mathbf{v}_1, \quad \mathbf{q}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|},$$

and, for $j = 2, \dots, k$,

$$\mathbf{u}_j = \mathbf{v}_j - \sum_{i=1}^{j-1} (\mathbf{q}_i' \mathbf{v}_j) \mathbf{q}_i, \quad \mathbf{q}_j = \frac{\mathbf{u}_j}{\|\mathbf{u}_j\|}.$$

Then every $\mathbf{u}_j$ is nonzero, $\mathbf{q}_1, \dots, \mathbf{q}_k$ are orthonormal, and

$$\text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_j\} = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_j\}, \quad j = 1, \dots, k.$$

Origin: H Thm. 6.4.1 and Cor. 6.4.2

Proof. For $j = 1$, $\mathbf{u}_1 \neq \mathbf{0}$,

$$\|\mathbf{q}_1\| = \frac{\|\mathbf{u}_1\|}{\|\mathbf{u}_1\|} = 1, \quad \text{span}\{\mathbf{q}_1\} = \text{span}\{\mathbf{v}_1\}.$$

Suppose the claims hold through $j - 1$. If $\mathbf{u}_j = \mathbf{0}$, then

$$\mathbf{v}_j = \sum_{i=1}^{j-1} (\mathbf{q}'_i \mathbf{v}_j) \mathbf{q}_i \in \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_{j-1}\},$$

contrary to independence. Thus $\|\mathbf{u}_j\| > 0$. For $1 \leq \ell < j$,

$$\mathbf{q}'_\ell \mathbf{u}_j = \mathbf{q}'_\ell \mathbf{v}_j - \sum_{i=1}^{j-1} (\mathbf{q}'_i \mathbf{v}_j) \mathbf{q}'_\ell \mathbf{q}_i = \mathbf{q}'_\ell \mathbf{v}_j - \mathbf{q}'_\ell \mathbf{v}_j = 0,$$

and $\mathbf{q}'_j \mathbf{q}_j = 1$. Moreover,

$$\mathbf{u}_j \in \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_j\}, \quad \mathbf{v}_j = \mathbf{u}_j + \sum_{i=1}^{j-1} (\mathbf{q}'_i \mathbf{v}_j) \mathbf{q}_i.$$

The two inclusions give the span equality for j . □

Corollary 6.5 (Existence and extension of orthonormal bases). Every subspace of $\mathbb{R}^n$ has an orthonormal basis. If an orthonormal list is contained in a subspace $\mathcal{S}$, then it extends to an orthonormal basis of $\mathcal{S}$.

Origin: H Thms. 6.4.3 and 6.4.5

Proof. The empty list is a basis of $\{\mathbf{0}\}$. For a nonzero subspace $\mathcal{S}$, apply Gram–Schmidt to any basis of $\mathcal{S}$.

Let $\mathbf{q}_1, \dots, \mathbf{q}_k \in \mathcal{S}$ be orthonormal. It is independent, so the basis-extension theorem gives a basis

$$\mathbf{q}_1, \dots, \mathbf{q}_k, \mathbf{v}_{k+1}, \dots, \mathbf{v}_d$$

of $\mathcal{S}$, where $d = \dim \mathcal{S}$. Gram–Schmidt leaves the first k vectors unchanged because

$$\mathbf{q}_j - \sum_{i=1}^{j-1} (\mathbf{q}'_i \mathbf{q}_j) \mathbf{q}_i = \mathbf{q}_j, \quad \|\mathbf{q}_j\| = 1.$$

Its output is the required extension. □

1.2 Orthonormal coordinates

Theorem 6.6 (Orthonormal coordinates and Parseval's identity). Let $\mathbf{q}_1, \dots, \mathbf{q}_k$ be an orthonormal basis of $\mathcal{S} \subseteq \mathbb{R}^n$. For every $\mathbf{x} \in \mathcal{S}$,

$$\mathbf{x} = \sum_{j=1}^k (\mathbf{q}'_j \mathbf{x}) \mathbf{q}_j, \quad \|\mathbf{x}\|^2 = \sum_{j=1}^k |\mathbf{q}'_j \mathbf{x}|^2.$$

If $k > 0$, $\mathbf{Q} = (\mathbf{q}_1, \dots, \mathbf{q}_k)$, and $\mathbf{x} = \mathbf{Q}\mathbf{c}$, then, equivalently,

$$\mathbf{c} = \mathbf{Q}'\mathbf{x}, \quad \|\mathbf{x}\| = \|\mathbf{c}\|.$$

Origin: H Thm. 6.4.4

Proof. If $k = 0$, then $\mathcal{S} = \{\mathbf{0}\}$ and both displayed identities are the empty-sum identity $0 = 0$.

Suppose $k > 0$. Write $\mathbf{x} = \sum_{j=1}^k c_j \mathbf{q}_j$. For each i ,

$$\mathbf{q}'_i \mathbf{x} = \sum_{j=1}^k c_j \mathbf{q}'_i \mathbf{q}_j = c_i.$$

Therefore $\mathbf{c} = \mathbf{Q}'\mathbf{x}$, and

$$\|\mathbf{x}\|^2 = \mathbf{c}'\mathbf{Q}'\mathbf{Q}\mathbf{c} = \mathbf{c}'\mathbf{c} = \sum_{j=1}^k |\mathbf{q}'_j \mathbf{x}|^2.$$

□

Example 6.1 (Exact orthonormalisation). For

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix},$$

Gram–Schmidt gives

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{u}_2 = \mathbf{v}_2 - (\mathbf{q}_1' \mathbf{v}_2) \mathbf{q}_1 = \begin{pmatrix} 1/2 \\ -1/2 \\ 1 \end{pmatrix}, \quad \mathbf{q}_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}.$$

Thus $\mathbf{Q} = (\mathbf{q}_1, \mathbf{q}_2)$ satisfies $\mathbf{Q}'\mathbf{Q} = \mathbf{I}_2$ and has the same column space as $(\mathbf{v}_1, \mathbf{v}_2)$.

Origin: Course formulation from H Thm. 6.4.1

1.3 Thin QR factorisation

Theorem 6.7 (Thin QR factorisation). Let $\mathbf{A} \in \mathbb{R}^{m \times p}$ have rank p , where $1 \leq p \leq m$. There are unique matrices $\mathbf{Q} \in \mathbb{R}^{m \times p}$ and $\mathbf{R} \in \mathbb{R}^{p \times p}$ such that

$$\mathbf{A} = \mathbf{QR}, \quad \mathbf{Q}'\mathbf{Q} = \mathbf{I}_p, \quad \mathbf{R} \text{ is upper triangular,} \quad r_{jj} > 0.$$

Origin: H §6.4d; A&M Exs. 7.34–7.35

Proof. Write $\mathbf{A} = (\mathbf{a}_1, \dots, \mathbf{a}_p)$ and apply Gram–Schmidt to its independent columns. Let $\mathbf{Q} = (\mathbf{q}_1, \dots, \mathbf{q}_p)$ and $\mathbf{R} = \mathbf{Q}'\mathbf{A}$. Since $\mathbf{a}_j \in \text{col}(\mathbf{Q})$, orthonormal coordinates give

$$\mathbf{a}_j = \sum_{i=1}^p (\mathbf{q}'_i \mathbf{a}_j) \mathbf{q}_i, \quad \mathbf{A} = \mathbf{Q}\mathbf{R}.$$

For $i > j$, $\mathbf{a}_j \in \text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_{i-1}\}$, so $r_{ij} = \mathbf{q}'_i \mathbf{a}_j = 0$. If $\mathbf{u}_j$ is the j th Gram–Schmidt residual, then

$$r_{jj} = \mathbf{q}'_j \mathbf{a}_j = \mathbf{q}'_j \mathbf{u}_j = \|\mathbf{u}_j\| > 0.$$

For uniqueness, the first column satisfies

$$\mathbf{a}_1 = r_{11} \mathbf{q}_1, \quad r_{11} = \|\mathbf{a}_1\| > 0, \quad \mathbf{q}_1 = \mathbf{a}_1 / r_{11}.$$

Suppose columns $\mathbf{q}_1, \dots, \mathbf{q}_{j-1}$ have been determined. From $\mathbf{a}_j = \sum_{i=1}^j r_{ij} \mathbf{q}_i$,

$$r_{ij} = \mathbf{q}'_i \mathbf{a}_j \quad (i < j), \quad \mathbf{w}_j = \mathbf{a}_j - \sum_{i=1}^{j-1} r_{ij} \mathbf{q}_i = r_{jj} \mathbf{q}_j.$$

Hence $r_{jj} = \|\mathbf{w}_j\| > 0$ and $\mathbf{q}_j = \mathbf{w}_j / r_{jj}$. Induction determines $\mathbf{Q}$ and $\mathbf{R}$. □

Example 6.2 (An exact thin QR factorisation). The vectors in Example 6.1 give

$$\begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{2} & -1/\sqrt{6} \\ 0 & 2/\sqrt{6} \end{pmatrix} \begin{pmatrix} \sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{pmatrix}.$$

The left factor has orthonormal columns; the right factor is upper triangular with positive diagonal.

Origin: Course formulation from H §6.4d

2 Orthogonal complements and projection

2.1 Complements and the four fundamental subspaces

Definition 6.8 (Orthogonal complement). For $\mathcal{S} \subseteq \mathbb{R}^n$, its **orthogonal complement** is

$$\mathcal{S}^\perp = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x}'\mathbf{s} = 0 \text{ for every } \mathbf{s} \in \mathcal{S}\}.$$

Origin: H §12.5a; S §2.3.1

Lemma 6.9 (Complement subspace and zero intersection). For every $\mathcal{S} \subseteq \mathbb{R}^n$, $\mathcal{S}^\perp$ is a subspace. If $\mathcal{S}$ is a subspace, then

$$\mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}.$$

Origin: S Prop. 2.4

Proof. For every $\mathbf{s} \in \mathcal{S}$,

$$\mathbf{0}'\mathbf{s} = 0,$$

so $\mathbf{0} \in \mathcal{S}^\perp$. For $\mathbf{x}, \mathbf{y} \in \mathcal{S}^\perp$, $a, b \in \mathbb{R}$, and $\mathbf{s} \in \mathcal{S}$,

$$(a\mathbf{x} + b\mathbf{y})'\mathbf{s} = a\mathbf{x}'\mathbf{s} + b\mathbf{y}'\mathbf{s} = 0.$$

Thus $a\mathbf{x} + b\mathbf{y} \in \mathcal{S}^\perp$. If $\mathbf{x} \in \mathcal{S} \cap \mathcal{S}^\perp$, then

$$\|\mathbf{x}\|^2 = \mathbf{x}'\mathbf{x} = 0,$$

so $\mathbf{x} = \mathbf{0}$.



Theorem 6.10 (Orthogonal decomposition). For every subspace $\mathcal{S} \subseteq \mathbb{R}^n$,

$$\mathbb{R}^n = \mathcal{S} \oplus \mathcal{S}^\perp, \quad \dim \mathcal{S}^\perp = n - \dim \mathcal{S}, \quad (\mathcal{S}^\perp)^\perp = \mathcal{S}.$$

Thus every $\mathbf{y} \in \mathbb{R}^n$ has unique $\mathbf{s} \in \mathcal{S}$ and $\mathbf{t} \in \mathcal{S}^\perp$ such that $\mathbf{y} = \mathbf{s} + \mathbf{t}$.

Origin: H Thms. 12.5.4, 12.5.11, and 12.5.12; S Props. 2.4–2.5

Proof. Let $\mathbf{q}_1, \dots, \mathbf{q}_k$ be an orthonormal basis of $\mathcal{S}$ and extend it to an orthonormal basis $\mathbf{q}_1, \dots, \mathbf{q}_n$ of $\mathbb{R}^n$. For $\mathbf{y} = \sum_{j=1}^n c_j \mathbf{q}_j$, put

$$\mathbf{s} = \sum_{j=1}^k c_j \mathbf{q}_j, \quad \mathbf{t} = \sum_{j=k+1}^n c_j \mathbf{q}_j.$$

Then $\mathbf{s} \in \mathcal{S}$ and, for $1 \leq i \leq k$,

$$\mathbf{q}_i' \mathbf{t} = \sum_{j=k+1}^n c_j \mathbf{q}_i' \mathbf{q}_j = 0,$$

so $\mathbf{t} \in \mathcal{S}^\perp$. If also $\mathbf{y} = \mathbf{s}_1 + \mathbf{t}_1$, then

$$\mathbf{s} - \mathbf{s}_1 = \mathbf{t}_1 - \mathbf{t} \in \mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}.$$

Hence the sum is direct and the decomposition is unique. The list $\mathbf{q}_{k+1}, \dots, \mathbf{q}_n$ is an orthonormal basis of $\mathcal{S}^\perp$, giving $\dim \mathcal{S}^\perp = n - k$. Since

$$\mathcal{S} \subseteq (\mathcal{S}^\perp)^\perp, \quad \dim(\mathcal{S}^\perp)^\perp = n - (n - k) = k = \dim \mathcal{S},$$

the two subspaces are equal. □

Theorem 6.11 (Four fundamental subspaces). For $\mathbf{A} \in \mathbb{R}^{m \times n}$,

$$\text{col}(\mathbf{A})^\perp = \text{null}(\mathbf{A}'), \quad \text{row}(\mathbf{A})^\perp = \text{null}(\mathbf{A}).$$

Consequently,

$$\mathbb{R}^m = \text{col}(\mathbf{A}) \oplus \text{null}(\mathbf{A}'), \quad \mathbb{R}^n = \text{row}(\mathbf{A}) \oplus \text{null}(\mathbf{A}).$$

Origin: S Prop. 2.6

Proof. Writing $\mathbf{A} = (\mathbf{a}_1, \dots, \mathbf{a}_n)$, for $\mathbf{y} \in \mathbb{R}^m$,

$$\begin{aligned}\mathbf{y} \in \text{col}(\mathbf{A})^\perp &\iff \mathbf{y}'\mathbf{a}_j = 0 \quad (j = 1, \dots, n) \\ &\iff \mathbf{A}'\mathbf{y} = \mathbf{0} \iff \mathbf{y} \in \text{null}(\mathbf{A}').\end{aligned}$$

Apply this identity to $\mathbf{A}'$ and use $\text{row}(\mathbf{A}) = \text{col}(\mathbf{A}')$:

$$\text{row}(\mathbf{A})^\perp = \text{col}(\mathbf{A}')^\perp = \text{null}(\mathbf{A}).$$

The two direct sums are the orthogonal-decomposition theorem applied to $\text{col}(\mathbf{A})$ and $\text{row}(\mathbf{A})$. $\square$

2.2 Projection and best approximation

Definition 6.12 (Orthogonal projection). Let $\mathcal{S} \subseteq \mathbb{R}^n$ be a subspace. If

$$\mathbf{y} = \mathbf{s} + \mathbf{t}, \quad \mathbf{s} \in \mathcal{S}, \quad \mathbf{t} \in \mathcal{S}^\perp,$$

then the **orthogonal projection** of $\mathbf{y}$ onto $\mathcal{S}$ is $P_{\mathcal{S}}\mathbf{y} = \mathbf{s}$. The map $P_{\mathcal{S}}$ is the **orthogonal projector** onto $\mathcal{S}$.

Origin: H §12.1

Theorem 6.13 (Matrix of an orthogonal projection). Let $\mathcal{S} \subseteq \mathbb{R}^n$ have dimension k . If $k > 0$, let $\mathbf{Q} \in \mathbb{R}^{n \times k}$ have orthonormal columns spanning $\mathcal{S}$. Then

$$P_{\mathcal{S}}\mathbf{y} = \mathbf{Q}\mathbf{Q}'\mathbf{y}, \quad \mathbf{P}_{\mathcal{S}} = \mathbf{Q}\mathbf{Q}'.$$

If $k = 0$, then $\mathbf{P}_{\mathcal{S}} = \mathbf{O}$. In both cases,

$$\mathbf{M}_{\mathcal{S}} = \mathbf{I}_n - \mathbf{P}_{\mathcal{S}} = \mathbf{P}_{\mathcal{S}^\perp}.$$

The map $P_{\mathcal{S}}$ is linear, and

$$\text{col}(\mathbf{P}_{\mathcal{S}}) = \mathcal{S}, \quad \text{null}(\mathbf{P}_{\mathcal{S}}) = \mathcal{S}^\perp, \quad \mathbf{P}'_{\mathcal{S}} = \mathbf{P}_{\mathcal{S}}, \quad \mathbf{P}_{\mathcal{S}}^2 = \mathbf{P}_{\mathcal{S}},$$

$$\mathbf{M}'_{\mathcal{S}} = \mathbf{M}_{\mathcal{S}}, \quad \mathbf{M}_{\mathcal{S}}^2 = \mathbf{M}_{\mathcal{S}}, \quad \mathbf{P}_{\mathcal{S}}\mathbf{M}_{\mathcal{S}} = \mathbf{M}_{\mathcal{S}}\mathbf{P}_{\mathcal{S}} = \mathbf{0}.$$

The matrix $\mathbf{P}_{\mathcal{S}}$ is independent of the orthonormal basis used for $\mathcal{S}$.

Origin: H Thms. 12.1.3–12.1.4 and 12.5.8

Proof. If $k = 0$, then $\mathcal{S} = \{\mathbf{0}\}$ and the orthogonal decomposition is $\mathbf{y} = \mathbf{0} + \mathbf{y}$, giving $\mathbf{P}_{\mathcal{S}} = \mathbf{0}$ and $\mathbf{M}_{\mathcal{S}} = \mathbf{I}_n$. Moreover,

$$\text{col}(\mathbf{0}) = \{\mathbf{0}\}, \quad \text{null}(\mathbf{0}) = \mathbb{R}^n, \quad \mathbf{0}' = \mathbf{0}^2 = \mathbf{0}, \quad \mathbf{I}'_n = \mathbf{I}_n^2 = \mathbf{I}_n, \quad \mathbf{0}\mathbf{I}_n = \mathbf{I}_n\mathbf{0} = \mathbf{0}.$$

Suppose $k > 0$. Orthonormal coordinates give

$$\mathbf{s} = \sum_{j=1}^k (\mathbf{q}'_j \mathbf{y}) \mathbf{q}_j = \mathbf{Q}\mathbf{Q}'\mathbf{y} \in \mathcal{S}.$$

For $\mathbf{x} = \mathbf{Q}\mathbf{c} \in \mathcal{S}$,

$$\mathbf{x}'(\mathbf{y} - \mathbf{s}) = \mathbf{c}'\mathbf{Q}'(\mathbf{I}_n - \mathbf{Q}\mathbf{Q}')\mathbf{y} = \mathbf{c}'(\mathbf{Q}' - \mathbf{Q}')\mathbf{y} = 0.$$

Thus $\mathbf{y} - \mathbf{s} \in \mathcal{S}^\perp$, so uniqueness of the orthogonal decomposition gives $P_{\mathcal{S}}\mathbf{y} = \mathbf{Q}\mathbf{Q}'\mathbf{y}$. Matrix

multiplication gives linearity and

$$(\mathbf{Q}\mathbf{Q}')' = \mathbf{Q}\mathbf{Q}', \quad (\mathbf{Q}\mathbf{Q}')^2 = \mathbf{Q}(\mathbf{Q}'\mathbf{Q})\mathbf{Q}' = \mathbf{Q}\mathbf{Q}'.$$

Moreover,

$$\mathbf{P}_{\mathcal{S}}\mathbf{x} = \mathbf{x} \quad (\mathbf{x} \in \mathcal{S}), \quad \mathbf{P}_{\mathcal{S}}\mathbf{x} = \mathbf{0} \iff \mathbf{Q}'\mathbf{x} = \mathbf{0} \iff \mathbf{x} \in \mathcal{S}^{\perp}.$$

Hence $\text{col}(\mathbf{P}_{\mathcal{S}}) = \mathcal{S}$ and $\text{null}(\mathbf{P}_{\mathcal{S}}) = \mathcal{S}^{\perp}$. Since

$$\mathbf{y} = P_{\mathcal{S}}\mathbf{y} + (\mathbf{I}_n - \mathbf{P}_{\mathcal{S}})\mathbf{y}$$

is its orthogonal decomposition, $\mathbf{I}_n - \mathbf{P}_{\mathcal{S}}$ projects onto $\mathcal{S}^{\perp}$. From $\mathbf{P}'_{\mathcal{S}} = \mathbf{P}_{\mathcal{S}}^2 = \mathbf{P}_{\mathcal{S}}$,

$$\mathbf{M}'_{\mathcal{S}} = \mathbf{I}_n - \mathbf{P}'_{\mathcal{S}} = \mathbf{M}_{\mathcal{S}}, \quad \mathbf{M}_{\mathcal{S}}^2 = \mathbf{I}_n - 2\mathbf{P}_{\mathcal{S}} + \mathbf{P}_{\mathcal{S}}^2 = \mathbf{M}_{\mathcal{S}},$$

$$\mathbf{P}_{\mathcal{S}}\mathbf{M}_{\mathcal{S}} = \mathbf{P}_{\mathcal{S}} - \mathbf{P}_{\mathcal{S}}^2 = \mathbf{0}, \quad \mathbf{M}_{\mathcal{S}}\mathbf{P}_{\mathcal{S}} = \mathbf{P}_{\mathcal{S}} - \mathbf{P}_{\mathcal{S}}^2 = \mathbf{0}.$$

Uniqueness of $P_{\mathcal{S}}\mathbf{y}$ for every $\mathbf{y}$ makes its matrix independent of the chosen basis. □

Theorem 6.14 (Best approximation). For $\mathbf{y} \in \mathbb{R}^n$ and a subspace $\mathcal{S} \subseteq \mathbb{R}^n$, $P_{\mathcal{S}}\mathbf{y}$ is the unique minimiser of $\|\mathbf{y} - \mathbf{s}\|$ over $\mathbf{s} \in \mathcal{S}$. For every $\mathbf{s} \in \mathcal{S}$,

$$\|\mathbf{y} - \mathbf{s}\|^2 = \|\mathbf{y} - P_{\mathcal{S}}\mathbf{y}\|^2 + \|P_{\mathcal{S}}\mathbf{y} - \mathbf{s}\|^2.$$

Origin: H Thm. 12.4.1

Proof. The vectors $\mathbf{y} - P_{\mathcal{S}}\mathbf{y} \in \mathcal{S}^\perp$ and $P_{\mathcal{S}}\mathbf{y} - \mathbf{s} \in \mathcal{S}$ are orthogonal. Therefore

$$\|\mathbf{y} - \mathbf{s}\|^2 = \|(\mathbf{y} - P_{\mathcal{S}}\mathbf{y}) + (P_{\mathcal{S}}\mathbf{y} - \mathbf{s})\|^2 = \|\mathbf{y} - P_{\mathcal{S}}\mathbf{y}\|^2 + \|P_{\mathcal{S}}\mathbf{y} - \mathbf{s}\|^2.$$

The second term is nonnegative and is zero if and only if $\mathbf{s} = P_{\mathcal{S}}\mathbf{y}$. □

Theorem 6.15 (Characterisation of orthogonal projectors). A matrix $\mathbf{P} \in \mathbb{R}^{n \times n}$ is the matrix of an orthogonal projection if and only if

$$\mathbf{P}' = \mathbf{P}, \quad \mathbf{P}^2 = \mathbf{P}.$$

In that case it projects onto $\text{col}(\mathbf{P})$ and

$$\text{null}(\mathbf{P}) = \text{col}(\mathbf{P})^\perp.$$

Origin: H Thm. 12.3.9

Proof. Every orthogonal projection matrix is symmetric and idempotent by Theorem 6.13. Conversely, suppose $\mathbf{P}' = \mathbf{P}^2 = \mathbf{P}$. For every $\mathbf{x}$,

$$\mathbf{x} = \mathbf{P}\mathbf{x} + (\mathbf{I}_n - \mathbf{P})\mathbf{x}, \quad \mathbf{P}\mathbf{x} \in \text{col}(\mathbf{P}),$$

and, for $\mathbf{s} = \mathbf{P}\mathbf{a} \in \text{col}(\mathbf{P})$,

$$\mathbf{s}'(\mathbf{I}_n - \mathbf{P})\mathbf{x} = \mathbf{a}'\mathbf{P}'(\mathbf{I}_n - \mathbf{P})\mathbf{x} = \mathbf{a}'(\mathbf{P} - \mathbf{P}^2)\mathbf{x} = 0.$$

Thus $(\mathbf{I}_n - \mathbf{P})\mathbf{x} \in \text{col}(\mathbf{P})^\perp$, so $\mathbf{P}\mathbf{x}$ is the orthogonal projection of $\mathbf{x}$ onto $\text{col}(\mathbf{P})$. If $\mathbf{P}\mathbf{x} = \mathbf{0}$, then, for every $\mathbf{s} = \mathbf{P}\mathbf{a}$,

$$\mathbf{s}'\mathbf{x} = \mathbf{a}'\mathbf{P}'\mathbf{x} = \mathbf{a}'\mathbf{P}\mathbf{x} = 0,$$

so $\mathbf{x} \in \text{col}(\mathbf{P})^\perp$. Conversely, if $\mathbf{x} \in \text{col}(\mathbf{P})^\perp$, then

$$\|\mathbf{P}\mathbf{x}\|^2 = \mathbf{x}'\mathbf{P}'\mathbf{P}\mathbf{x} = \mathbf{x}'\mathbf{P}\mathbf{x} = 0,$$

so $\mathbf{P}\mathbf{x} = \mathbf{0}$. □

2.3 Full-rank formulas and complementary projectors

Theorem 6.16 (Full-column-rank projection formula). If $\mathbf{X} \in \mathbb{R}^{n \times p}$ has rank p , then

$$\mathbf{P}_{\text{col}(\mathbf{X})} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}', \quad \mathbf{M}_{\text{col}(\mathbf{X})} = \mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'.$$

If $\mathbf{X} = \mathbf{Q}\mathbf{R}$ is its thin QR factorisation, then

$$\mathbf{P}_{\text{col}(\mathbf{X})} = \mathbf{Q}\mathbf{Q}'.$$

Origin: H Cor. 12.2.3; CLM Thms. 2.2–2.3

Proof. Since $\mathbf{R}$ is nonsingular,

$$\text{col}(\mathbf{X}) = \text{col}(\mathbf{QR}) = \text{col}(\mathbf{Q}), \quad \mathbf{X}'\mathbf{X} = \mathbf{R}'\mathbf{Q}'\mathbf{QR} = \mathbf{R}'\mathbf{R}.$$

Hence

$$\begin{aligned} \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' &= \mathbf{QR}(\mathbf{R}'\mathbf{R})^{-1}\mathbf{R}'\mathbf{Q}' \\ &= \mathbf{QRR}^{-1}(\mathbf{R}')^{-1}\mathbf{R}'\mathbf{Q}' = \mathbf{QQ}'. \end{aligned}$$

The complementary formula is $\mathbf{M}_{\mathcal{S}} = \mathbf{I}_n - \mathbf{P}_{\mathcal{S}}$. □

Theorem 6.17 (Trace and rank of an orthogonal projector). If $\mathbf{P}$ is the orthogonal projector onto a k -dimensional subspace of $\mathbb{R}^n$, then

$$\text{rank}(\mathbf{P}) = \text{tr}(\mathbf{P}) = k.$$

Origin: Adapted from H Cor. 10.2.2 and §12.1

Proof. If $k = 0$, then $\mathbf{P} = \mathbf{O}$, so $\text{rank}(\mathbf{P}) = \text{tr}(\mathbf{P}) = 0$. Suppose $k > 0$. Let $\mathbf{Q} \in \mathbb{R}^{n \times k}$ have orthonormal columns spanning the subspace. Then $\mathbf{P} = \mathbf{Q}\mathbf{Q}'$ and $\text{col}(\mathbf{P}) = \text{col}(\mathbf{Q})$, so

$$\text{rank}(\mathbf{P}) = \dim \text{col}(\mathbf{P}) = k.$$

The cyclic trace identity gives

$$\text{tr}(\mathbf{P}) = \text{tr}(\mathbf{Q}\mathbf{Q}') = \text{tr}(\mathbf{Q}'\mathbf{Q}) = \text{tr}(\mathbf{I}_k) = k.$$

□

Example 6.3 (Mean and centred components). For $\mathcal{S} = \text{span}\{\mathbf{1}\} \subseteq \mathbb{R}^n$,

$$\mathbf{q} = \frac{1}{\sqrt{n}}\mathbf{1}, \quad \mathbf{P}_{\mathcal{S}} = \frac{1}{n}\mathbf{1}\mathbf{1}', \quad \mathbf{M}_{\mathcal{S}} = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}'.$$

For $\mathbf{y} = (y_1, \dots, y_n)'$ and $\bar{y} = n^{-1} \sum_i y_i$,

$$\mathbf{P}_{\mathcal{S}}\mathbf{y} = \bar{y}\mathbf{1}, \quad \mathbf{M}_{\mathcal{S}}\mathbf{y} = \mathbf{y} - \bar{y}\mathbf{1}, \quad \mathbf{1}'\mathbf{M}_{\mathcal{S}}\mathbf{y} = 0.$$

3 Least squares

3.1 Normal equations and all minimisers

Definition 6.18 (Least-squares problem). For $\mathbf{X} \in \mathbb{R}^{n \times p}$ and $\mathbf{y} \in \mathbb{R}^n$, a **least-squares minimiser** is any $\hat{\boldsymbol{\beta}} \in \mathbb{R}^p$ satisfying

$$\|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 = \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2.$$

Its fitted vector and residual vector are

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}, \quad \hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \hat{\mathbf{y}}.$$

Origin: H §12.4; CLM §2.1

Theorem 6.19 (Normal equations and the minimiser set). For arbitrary $\mathbf{X} \in \mathbb{R}^{n \times p}$, the normal equations

$$\mathbf{X}'\mathbf{X}\boldsymbol{\beta} = \mathbf{X}'\mathbf{y}$$

are consistent. For $\boldsymbol{\beta}_0 \in \mathbb{R}^p$, the following are equivalent:

- (i) $\boldsymbol{\beta}_0$ is a least-squares minimiser;
- (ii) $\mathbf{X}\boldsymbol{\beta}_0 = P_{\text{col}(\mathbf{X})}\mathbf{y}$;
- (iii) $\mathbf{y} - \mathbf{X}\boldsymbol{\beta}_0 \in \text{null}(\mathbf{X}')$;
- (iv) $\mathbf{X}'\mathbf{X}\boldsymbol{\beta}_0 = \mathbf{X}'\mathbf{y}$.

If $\boldsymbol{\beta}_0$ is one minimiser, then all minimisers form

$$\boldsymbol{\beta}_0 + \text{null}(\mathbf{X}).$$

The fitted vector and residual vector are unique. The coefficient minimiser is unique if and only if $\text{rank}(\mathbf{X}) = p$.

Origin: H Thm. 7.4.1, Cors. 12.2.3–12.2.4, and Thms. 12.4.2–12.4.3

Proof. Put $\mathcal{S} = \text{col}(\mathbf{X})$ and $\hat{\mathbf{y}} = P_{\mathcal{S}}\mathbf{y}$. Since $\hat{\mathbf{y}} \in \text{col}(\mathbf{X})$, choose $\boldsymbol{\beta}_*$ with $\mathbf{X}\boldsymbol{\beta}_* = \hat{\mathbf{y}}$. For every $\boldsymbol{\beta} \in \mathbb{R}^p$,

$$\mathbf{y} - \mathbf{X}\boldsymbol{\beta} = (\mathbf{y} - \hat{\mathbf{y}}) + (\hat{\mathbf{y}} - \mathbf{X}\boldsymbol{\beta}),$$

where the two terms lie in $\mathcal{S}^\perp$ and $\mathcal{S}$, respectively. Hence

$$\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \|\mathbf{y} - \hat{\mathbf{y}}\|^2 + \|\hat{\mathbf{y}} - \mathbf{X}\boldsymbol{\beta}\|^2.$$

Thus $\boldsymbol{\beta}$ minimises the objective if and only if $\mathbf{X}\boldsymbol{\beta} = \hat{\mathbf{y}}$. Taking $\boldsymbol{\beta} = \boldsymbol{\beta}_*$ proves existence; taking $\boldsymbol{\beta} = \boldsymbol{\beta}_0$ proves (i) $\iff$ (ii). The four-fundamental-subspaces theorem gives

$$\begin{aligned} \mathbf{X}\boldsymbol{\beta} = \hat{\mathbf{y}} &\iff \mathbf{y} - \mathbf{X}\boldsymbol{\beta} \in \text{col}(\mathbf{X})^\perp \\ &\iff \mathbf{y} - \mathbf{X}\boldsymbol{\beta} \in \text{null}(\mathbf{X}') \\ &\iff \mathbf{X}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{0}, \end{aligned}$$

which proves the remaining equivalences. If $\boldsymbol{\beta}_0$ is a minimiser, then

$$\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta}_0 \iff \mathbf{X}(\boldsymbol{\beta} - \boldsymbol{\beta}_0) = \mathbf{0} \iff \boldsymbol{\beta} \in \boldsymbol{\beta}_0 + \text{null}(\mathbf{X}).$$

Every minimiser therefore has the same fit and residual, and it is unique exactly when

$$\text{null}(\mathbf{X}) = \{\mathbf{0}\} \iff \text{rank}(\mathbf{X}) = p.$$

□

Corollary 6.20 (Full-rank least squares by QR). Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ have rank p and thin QR factorisation $\mathbf{X} = \mathbf{QR}$. The unique least-squares minimiser is

$$\hat{\boldsymbol{\beta}} = \mathbf{R}^{-1}\mathbf{Q}'\mathbf{y} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

Moreover,

$$\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \|\mathbf{Q}'\mathbf{y} - \mathbf{R}\boldsymbol{\beta}\|^2 + \|(\mathbf{I}_n - \mathbf{Q}\mathbf{Q}')\mathbf{y}\|^2.$$

Origin: H §6.4d and Cor. 12.2.3; A&M Ex. 7.35

Proof. The orthogonal decomposition of $\mathbf{y}$ gives

$$\mathbf{y} - \mathbf{QR}\boldsymbol{\beta} = \mathbf{Q}(\mathbf{Q}'\mathbf{y} - \mathbf{R}\boldsymbol{\beta}) + (\mathbf{I}_n - \mathbf{Q}\mathbf{Q}')\mathbf{y}.$$

The summands lie in $\text{col}(\mathbf{Q})$ and $\text{col}(\mathbf{Q})^\perp$, so Parseval's identity and Pythagoras give the displayed norm identity. Its first term is zero exactly when

$$\mathbf{R}\boldsymbol{\beta} = \mathbf{Q}'\mathbf{y} \iff \boldsymbol{\beta} = \mathbf{R}^{-1}\mathbf{Q}'\mathbf{y}.$$

The full-column-rank projection formula gives

$$\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{Q}\mathbf{Q}'\mathbf{y} = \mathbf{X}\mathbf{R}^{-1}\mathbf{Q}'\mathbf{y}.$$

Cancellation by the full-column-rank matrix $\mathbf{X}$ proves the second coefficient formula. □

3.2 Hat, residual, and leverage

Definition 6.21 (Hat matrix, residual maker, and leverage). For $\mathbf{X} \in \mathbb{R}^{n \times p}$, define

$$\mathbf{H} = \mathbf{P}_{\text{col}(\mathbf{X})}, \quad \mathbf{M} = \mathbf{I}_n - \mathbf{H}.$$

The diagonal entry $h_{ii} = \mathbf{e}_i' \mathbf{H} \mathbf{e}_i$ is the **leverage** of row i .

Origin: CLM Thms. 2.3 and 2.5

Theorem 6.22 (Fitted and residual identities). For every least-squares minimiser,

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \quad \hat{\boldsymbol{\varepsilon}} = \mathbf{M}\mathbf{y}, \quad \mathbf{X}'\hat{\boldsymbol{\varepsilon}} = \mathbf{0}, \quad \mathbf{H}\hat{\boldsymbol{\varepsilon}} = \mathbf{0}, \quad \mathbf{M}\hat{\mathbf{y}} = \mathbf{0},$$

and

$$\|\mathbf{y}\|^2 = \|\hat{\mathbf{y}}\|^2 + \|\hat{\boldsymbol{\varepsilon}}\|^2.$$

If $\mathbf{1} \in \text{col}(\mathbf{X})$, then $\mathbf{1}'\hat{\boldsymbol{\varepsilon}} = 0$.

Origin: CLM Thms. 2.3–2.4; H Thm. 12.2.5

Proof. The normal-equation theorem and complementary projection give

$$\hat{\mathbf{y}} = P_{\text{col}(\mathbf{X})}\mathbf{y} = \mathbf{H}\mathbf{y}, \quad \hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{M}\mathbf{y} \in \text{col}(\mathbf{X})^\perp = \text{null}(\mathbf{X}').$$

Thus $\mathbf{X}'\hat{\boldsymbol{\varepsilon}} = \mathbf{0}$. Since $\hat{\mathbf{y}} \in \text{col}(\mathbf{H})$ and $\hat{\boldsymbol{\varepsilon}} \in \text{null}(\mathbf{H}) = \text{col}(\mathbf{H})^\perp$,

$$\mathbf{H}\hat{\boldsymbol{\varepsilon}} = \mathbf{0}, \quad \mathbf{M}\hat{\mathbf{y}} = (\mathbf{I}_n - \mathbf{H})\hat{\mathbf{y}} = \mathbf{0}, \quad \hat{\mathbf{y}}'\hat{\boldsymbol{\varepsilon}} = 0.$$

Pythagoras yields the norm identity. If $\mathbf{1} \in \text{col}(\mathbf{X})$, then $\hat{\boldsymbol{\varepsilon}} \perp \mathbf{1}$, so $\mathbf{1}'\hat{\boldsymbol{\varepsilon}} = 0$. □

Theorem 6.23 (Leverage identities and bounds). Let $r = \text{rank}(\mathbf{X})$. If $r > 0$, let $\mathbf{Q} \in \mathbb{R}^{n \times r}$ have

orthonormal columns spanning $\text{col}(\mathbf{X})$. Then

$$h_{ii} = \sum_{j=1}^r q_{ij}^2, \quad 0 \leq h_{ii} \leq 1, \quad \sum_{i=1}^n h_{ii} = r.$$

For $r = 0$, the sum defining h_{ii} is empty and equals zero. Moreover,

$$h_{ii} = 1 \iff \mathbf{e}_i \in \text{col}(\mathbf{X}), \quad h_{ii} = 0 \iff \mathbf{e}_i \in \text{col}(\mathbf{X})^\perp.$$

Writing $\hat{\boldsymbol{\varepsilon}} = (\hat{\varepsilon}_1, \dots, \hat{\varepsilon}_n)'$, if $h_{ii} = 1$, then $\hat{\varepsilon}_i = 0$ for every response $\mathbf{y}$.

Origin: CLM Thm. 2.5; H Cor. 10.2.2

Proof. If $r = 0$, then $\text{col}(\mathbf{X}) = \{\mathbf{0}\}$ and $\mathbf{H} = \mathbf{O}$, so

$$h_{ii} = 0, \quad \sum_{i=1}^n h_{ii} = 0, \quad \mathbf{e}_i \notin \{\mathbf{0}\}, \quad \mathbf{e}_i \in \mathbb{R}^n = \{\mathbf{0}\}^\perp.$$

Suppose $r > 0$. Since $\mathbf{H} = \mathbf{Q}\mathbf{Q}'$,

$$h_{ii} = \mathbf{e}_i' \mathbf{Q}\mathbf{Q}' \mathbf{e}_i = \|\mathbf{Q}' \mathbf{e}_i\|^2 = \sum_{j=1}^r q_{ij}^2 = \|\mathbf{H}\mathbf{e}_i\|^2 \geq 0.$$

For $\mathbf{M} = \mathbf{I}_n - \mathbf{H}$,

$$1 - h_{ii} = \mathbf{e}_i' \mathbf{M}\mathbf{e}_i = \|\mathbf{M}\mathbf{e}_i\|^2 \geq 0.$$

The trace-rank theorem gives $\sum_i h_{ii} = \text{tr}(\mathbf{H}) = r$. Also,

$$h_{ii} = 1 \iff \mathbf{M}\mathbf{e}_i = \mathbf{0} \iff \mathbf{e}_i \in \text{null}(\mathbf{M}) = \text{col}(\mathbf{X}),$$

and

$$h_{ii} = 0 \iff \mathbf{H}\mathbf{e}_i = \mathbf{0} \iff \mathbf{e}_i \in \text{null}(\mathbf{H}) = \text{col}(\mathbf{X})^\perp.$$

If $h_{ii} = 1$, then

$$\hat{\varepsilon}_i = \mathbf{e}_i' \mathbf{M}\mathbf{y} = (\mathbf{M}\mathbf{e}_i)' \mathbf{y} = 0.$$

□

3.3 Frisch–Waugh–Lovell decomposition

Theorem 6.24 (Frisch–Waugh–Lovell). Let

$$\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2) \in \mathbb{R}^{n \times (p_1 + p_2)}$$

have full column rank, where $p_1, p_2 \geq 1$. Let $\mathbf{P}_2$ project onto $\text{col}(\mathbf{X}_2)$, $\mathbf{M}_2 = \mathbf{I}_n - \mathbf{P}_2$, and $\mathbf{Z} = \mathbf{M}_2 \mathbf{X}_1$. Write

$$\mathbf{P}_\mathbf{X} = \mathbf{P}_{\text{col}(\mathbf{X})}, \quad \mathbf{P}_\mathbf{Z} = \mathbf{P}_{\text{col}(\mathbf{Z})}.$$

Then $\mathbf{Z}$ has full column rank,

$$\text{col}(\mathbf{X}) = \text{col}(\mathbf{X}_2) \oplus \text{col}(\mathbf{Z}), \quad \mathbf{P}_\mathbf{X} = \mathbf{P}_2 + \mathbf{P}_\mathbf{Z},$$

and, for the unique least-squares coefficient

$$\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{\beta}}'_1, \hat{\boldsymbol{\beta}}'_2)',$$

the first block is

$$\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}'_1 \mathbf{M}_2 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{M}_2 \mathbf{y} = (\mathbf{Z}' \mathbf{Z})^{-1} \mathbf{Z}' \mathbf{M}_2 \mathbf{y}.$$

If $\mathbf{M}_\mathbf{X} = \mathbf{I}_n - \mathbf{P}_\mathbf{X}$ and $\mathbf{M}_\mathbf{Z} = \mathbf{I}_n - \mathbf{P}_\mathbf{Z}$, then

$$\mathbf{M}_\mathbf{X} = \mathbf{M}_\mathbf{Z}\mathbf{M}_2 = \mathbf{M}_2\mathbf{M}_\mathbf{Z}.$$

Origin: CLM Thm. 8.1; H Thm. 12.3.5

Proof. If $\mathbf{Za} = \mathbf{0}$, then

$$\mathbf{X}_1\mathbf{a} = \mathbf{P}_2\mathbf{X}_1\mathbf{a} \in \text{col}(\mathbf{X}_2).$$

Choose $\mathbf{c}$ with $\mathbf{X}_1\mathbf{a} = \mathbf{X}_2\mathbf{c}$. Full column rank of $\mathbf{X}$ gives

$$\mathbf{X} \begin{pmatrix} \mathbf{a} \\ -\mathbf{c} \end{pmatrix} = \mathbf{0} \implies \mathbf{a} = \mathbf{0},$$

so $\mathbf{Z}$ has full column rank. Moreover,

$$\mathbf{X}_2'\mathbf{Z} = \mathbf{X}_2'\mathbf{M}_2\mathbf{X}_1 = \mathbf{0},$$

and

$$\mathbf{X}_1\mathbf{a} + \mathbf{X}_2\mathbf{c} = \mathbf{Za} + (\mathbf{P}_2\mathbf{X}_1\mathbf{a} + \mathbf{X}_2\mathbf{c}).$$

Both summands on the right belong to $\text{col}(\mathbf{X})$; they belong respectively to $\text{col}(\mathbf{Z})$ and $\text{col}(\mathbf{X}_2)$. Thus

$$\text{col}(\mathbf{X}) = \text{col}(\mathbf{X}_2) \oplus \text{col}(\mathbf{Z}).$$

For every $\mathbf{y}$, $\mathbf{P}_2\mathbf{y} + \mathbf{P}_Z\mathbf{y}$ lies in this direct sum. For $\mathbf{s}_2 \in \text{col}(\mathbf{X}_2)$ and $\mathbf{z}_0 \in \text{col}(\mathbf{Z})$,

$$\mathbf{s}_2'(\mathbf{y} - \mathbf{P}_2\mathbf{y} - \mathbf{P}_Z\mathbf{y}) = \mathbf{s}_2'(\mathbf{y} - \mathbf{P}_2\mathbf{y}) - \mathbf{s}_2'\mathbf{P}_Z\mathbf{y} = 0,$$

$$\mathbf{z}_0'(\mathbf{y} - \mathbf{P}_2\mathbf{y} - \mathbf{P}_Z\mathbf{y}) = \mathbf{z}_0'(\mathbf{y} - \mathbf{P}_Z\mathbf{y}) - \mathbf{z}_0'\mathbf{P}_2\mathbf{y} = 0.$$

Hence $\mathbf{P}_X = \mathbf{P}_2 + \mathbf{P}_Z$.

Since $\mathbf{P}_2\mathbf{X}_1 \in \text{col}(\mathbf{X}_2)$, choose $\mathbf{C}$ such that $\mathbf{P}_2\mathbf{X}_1 = \mathbf{X}_2\mathbf{C}$. Then

$$\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 = \mathbf{Z}\boldsymbol{\beta}_1 + \mathbf{X}_2(\mathbf{C}\boldsymbol{\beta}_1 + \boldsymbol{\beta}_2).$$

The $\text{col}(\mathbf{Z})$ component of the fitted vector is $\mathbf{P}_Z\mathbf{y}$, so uniqueness of coordinates in the full-column-rank matrix $\mathbf{Z}$ gives

$$\widehat{\boldsymbol{\beta}}_1 = (\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}.$$

Because $\mathbf{M}'_2 = \mathbf{M}_2^2 = \mathbf{M}_2$,

$$\mathbf{Z}'\mathbf{Z} = \mathbf{X}'_1\mathbf{M}_2\mathbf{X}_1, \quad \mathbf{Z}'\mathbf{y} = \mathbf{X}'_1\mathbf{M}_2\mathbf{y} = \mathbf{Z}'\mathbf{M}_2\mathbf{y}.$$

Finally, $\mathbf{P}_2\mathbf{P}_Z = \mathbf{P}_Z\mathbf{P}_2 = \mathbf{O}$, and therefore

$$\mathbf{M}_X = \mathbf{I}_n - \mathbf{P}_2 - \mathbf{P}_Z = (\mathbf{I}_n - \mathbf{P}_Z)(\mathbf{I}_n - \mathbf{P}_2) = (\mathbf{I}_n - \mathbf{P}_2)(\mathbf{I}_n - \mathbf{P}_Z).$$



4 Computation: Numerical Checks in R

4.1 Gram–Schmidt and QR residuals

```
1 cgs <- function(A) {  
2   n <- nrow(A); p <- ncol(A)  
3   Q <- matrix(0, n, p); R <- matrix(0, p, p)  
4   for (j in seq_len(p)) {  
5     if (j == 1L) {  
6       v <- A[, j]  
7     } else {
```

```
8       R[seq_len(j - 1L), j] <- crossprod(Q[, seq_len(j - 1L), drop = FALSE], A[, j])
9       v <- A[, j] - Q[, seq_len(j - 1L), drop = FALSE] %*% R[seq_len(j - 1L), j]
10    }
11    R[j, j] <- sqrt(sum(v^2))
12    Q[, j] <- v / R[j, j]
13  }
14  list(Q = Q, R = R)
15 }
16
17 mgs <- function(A) {
18   n <- nrow(A); p <- ncol(A)
19   Q <- matrix(0, n, p); R <- matrix(0, p, p)
20   for (j in seq_len(p)) {
21     v <- A[, j]
22     if (j > 1L) {
23       for (i in seq_len(j - 1L)) {
24         R[i, j] <- sum(Q[, i] * v)
25         v <- v - R[i, j] * Q[, i]
26       }
27     }
28     R[j, j] <- sqrt(sum(v^2))
```

```
29   Q[, j] <- v / R[j, j]
30 }
31 list(Q = Q, R = R)
32 }
33
34 grid <- seq(0, 1, length.out = 80)
35 A <- outer(grid, 0:7, ```)
36 fac_cgs <- cgs(A)
37 fac_mgs <- mgs(A)
38 fac_qr <- qr(A, LAPACK = TRUE)
39 Q_qr <- qr.Q(fac_qr)
40 R_qr <- qr.R(fac_qr)
41 A_qr <- A[, fac_qr$pivot, drop = FALSE]
42 rbind(
43   classical = c(reconstruction = max(abs(A - fac_cgs$Q %*% fac_cgs$R)),
44               orthogonality = max(abs(crossprod(fac_cgs$Q) - diag(ncol(A))))),
45   modified = c(reconstruction = max(abs(A - fac_mgs$Q %*% fac_mgs$R)),
46               orthogonality = max(abs(crossprod(fac_mgs$Q) - diag(ncol(A))))),
47   package_qr = c(reconstruction = max(abs(A_qr - Q_qr %*% R_qr)),
48                 orthogonality = max(abs(crossprod(Q_qr) - diag(ncol(A)))))
```

49)

```
##           reconstruction orthogonality
## classical      2.220e-16      4.510e-07
## modified      2.220e-16      1.112e-12
## package_qr     8.882e-16      1.443e-15
```

4.2 Projection is basis-free

```
1 data(Boston, package = "ISLR2")
2 X <- cbind(intercept = 1, as.matrix(Boston[, c("lstat", "rm", "age")]))
3 H_formula <- X %*% solve(crossprod(X), t(X))
4 Q <- qr.Q(qr(X, LAPACK = TRUE))
5 H_qr <- tcrossprod(Q)
6 Xs <- X[, c(2, 1, 4, 3)] %*% diag(c(10, 1, 0.1, 5))
7 Qs <- qr.Q(qr(Xs, LAPACK = TRUE))
8 H_rescaled <- tcrossprod(Qs)
9 c(formula_vs_qr = max(abs(H_formula - H_qr)),
10   qr_vs_rescaled = max(abs(H_qr - H_rescaled)))
```

```
## formula_vs_qr qr_vs_rescaled
##      3.386e-15      1.785e-15
```

4.3 Normal equations and QR

```
1 y <- Boston$medv
2 b_ne <- solve(crossprod(X), crossprod(X, y))
3 b_qr <- qr.solve(X, y)
4 fit_ne <- as.vector(X %*% b_ne)
5 fit_qr <- as.vector(X %*% b_qr)
6 c(coefficient_difference = max(abs(b_ne - b_qr)),
7   fitted_difference = max(abs(fit_ne - fit_qr)),
8   normal_residual_normal_equations = max(abs(crossprod(X, y - fit_ne))),
9   normal_residual_qr = max(abs(crossprod(X, y - fit_qr))))
```

```
##           coefficient_difference
##                1.335e-12
##           fitted_difference
##                5.294e-13
## normal_residual_normal_equations
##                3.790e-10
##           normal_residual_qr
##                1.394e-10
```

4.4 FWL and leverage

```

1 X1 <- X[, "rm", drop = FALSE]
2 X2 <- X[, c("intercept", "lstat", "age"), drop = FALSE]
3 Q2 <- qr.Q(qr(X2, LAPACK = TRUE))
4 z <- X1 - Q2 %*% crossprod(Q2, X1)
5 y2 <- y - Q2 %*% crossprod(Q2, y)
6 b_fwl <- as.numeric(crossprod(z, y2) / crossprod(z))
7 b_full <- unname(b_qr[match("rm", colnames(X))])
8 res_full <- y - fit_qr
9 res_fwl <- as.vector(y2 - z * b_fwl)
10 h_qr <- rowSums(Q^2)
11 h_lm <- hatvalues(lm(medv ~ lstat + rm + age, data = Boston))
12 c(coefficient_difference = abs(b_full - b_fwl),
13   residual_difference = max(abs(res_full - res_fwl)),
14   leverage_difference = max(abs(h_qr - h_lm)),
15   residual_sum = abs(sum(res_full)),
16   normal_residual = max(abs(crossprod(X, res_full))))

```

```

## coefficient_difference    residual_difference
##                1.776e-15                3.642e-14

```

```
##      leverage_difference      residual_sum
##      5.482e-16                1.967e-12
##      normal_residual
##      1.394e-10
```

5 Pitfalls

5.1 Forming the full hat matrix for its diagonal

$$\mathbf{Q}'\mathbf{Q} = \mathbf{I}_r, \quad \text{col}(\mathbf{Q}) = \text{col}(\mathbf{X}), \quad \mathbf{H} = \mathbf{Q}\mathbf{Q}' \in \mathbb{R}^{n \times n}, \quad h_{ii} = \sum_{j=1}^r q_{ij}^2.$$

For $r \geq 1$, the row sums use nr multiplications and $n(r-1)$ additions. Forming $\mathbf{Q}\mathbf{Q}'$ by scalar products uses n^2r multiplications, $n^2(r-1)$ additions, and stores n^2 entries.

5.2 Confusing a projection matrix with an orthogonal matrix

$$\mathbf{P}' = \mathbf{P}, \quad \mathbf{P}^2 = \mathbf{P}, \quad \mathbf{P}'\mathbf{P} = \mathbf{I}_n \implies \mathbf{I}_n = \mathbf{P}'\mathbf{P} = \mathbf{P}^2 = \mathbf{P}.$$

Thus an orthogonal projection matrix is an orthogonal matrix only when it is $\mathbf{I}_n$.

5.3 Confusing a unique fit with unique coefficients

$$\hat{\boldsymbol{\beta}} + \mathbf{n}, \quad \mathbf{n} \in \text{null}(\mathbf{X}) \implies \mathbf{X}(\hat{\boldsymbol{\beta}} + \mathbf{n}) = \mathbf{X}\hat{\boldsymbol{\beta}}.$$

The fit is unique for every $\mathbf{X}$; the coefficient vector is unique exactly when $\text{null}(\mathbf{X}) = \{\mathbf{0}\}$.

6 Exercises

Exercise 6.1. Let $k \geq 1$ and $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$. Set $\mathcal{S}_0 = \{\mathbf{0}\}$ and $\mathcal{S}_j = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_j\}$. At step j , define

$$\mathbf{u}_j = \mathbf{v}_j - P_{\mathcal{S}_{j-1}} \mathbf{v}_j.$$

If $\mathbf{u}_j \neq \mathbf{0}$, retain $\mathbf{q} = \mathbf{u}_j / \|\mathbf{u}_j\|$; if $\mathbf{u}_j = \mathbf{0}$, retain no vector. Prove that the retained vectors are an orthonormal basis of $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$.

Origin: H Ex. 6.4

Solution. For every j ,

$$\mathbf{u}_j = (\mathbf{I}_n - \mathbf{P}_{\mathcal{S}_{j-1}}) \mathbf{v}_j \in \mathcal{S}_{j-1}^\perp, \quad \mathbf{v}_j = P_{\mathcal{S}_{j-1}} \mathbf{v}_j + \mathbf{u}_j.$$

If $\mathbf{u}_j = \mathbf{0}$, then $\mathbf{v}_j \in \mathcal{S}_{j-1}$ and $\mathcal{S}_j = \mathcal{S}_{j-1}$. If $\mathbf{u}_j \neq \mathbf{0}$, then the retained unit vector is orthogonal to

every earlier retained vector and

$$\mathcal{S}_j = \mathcal{S}_{j-1} \oplus \text{span}\{\mathbf{u}_j\}.$$

Induction gives an orthonormal basis of $\mathcal{S}_k = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$. ■

Exercise 6.2. Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have rank r , where $0 < r < n$. Prove that there are a permutation matrix $\mathbf{\Pi} \in \mathbb{R}^{n \times n}$, a matrix $\mathbf{Q} \in \mathbb{R}^{m \times r}$ with $\mathbf{Q}'\mathbf{Q} = \mathbf{I}_r$, and $\mathbf{R} = (\mathbf{R}_{11}, \mathbf{R}_{12}) \in \mathbb{R}^{r \times n}$ such that

$$\mathbf{A}\mathbf{\Pi} = \mathbf{Q}\mathbf{R},$$

where $\mathbf{R}_{11} \in \mathbb{R}^{r \times r}$ is upper triangular with positive diagonal.

Origin: H Ex. 6.5

Solution. Select r independent columns of $\mathbf{A}$ and permute them to the front:

$$\mathbf{A}\mathbf{\Pi} = (\mathbf{B}, \mathbf{C}), \quad \text{col}(\mathbf{B}) = \text{col}(\mathbf{A}).$$

The thin QR factorisation gives

$$\mathbf{B} = \mathbf{Q}\mathbf{R}_{11}, \quad \mathbf{Q}'\mathbf{Q} = \mathbf{I}_r,$$

with $\mathbf{R}_{11}$ upper triangular and positive on its diagonal. Every column of $\mathbf{C}$ belongs to $\text{col}(\mathbf{Q})$, so orthonormal coordinates give

$$\mathbf{C} = \mathbf{Q}\mathbf{Q}'\mathbf{C}.$$

Taking $\mathbf{R}_{12} = \mathbf{Q}'\mathbf{C}$ yields

$$\mathbf{A}\mathbf{\Pi} = (\mathbf{Q}\mathbf{R}_{11}, \mathbf{Q}\mathbf{R}_{12}) = \mathbf{Q}(\mathbf{R}_{11}, \mathbf{R}_{12}).$$

Exercise 6.3. In Gram–Schmidt, let

$$\mathbf{Q}_{j-1} = (\mathbf{q}_1, \dots, \mathbf{q}_{j-1}), \quad \mathcal{S}_{j-1} = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_{j-1}\}.$$

For $j = 2, \dots, k$, prove

$$\mathbf{u}_j = (\mathbf{I}_n - \mathbf{Q}_{j-1}\mathbf{Q}_{j-1}')\mathbf{v}_j = P_{\mathcal{S}_{j-1}^\perp} \mathbf{v}_j,$$

and

$$\|\mathbf{u}_j\| = \min_{\mathbf{s} \in \mathcal{S}_{j-1}} \|\mathbf{v}_j - \mathbf{s}\|.$$

Origin: H Ex. 12.8

Solution. The prefix-span identity gives $\text{col}(\mathbf{Q}_{j-1}) = \mathcal{S}_{j-1}$. Hence

$$P_{\mathcal{S}_{j-1}} = \mathbf{Q}_{j-1} \mathbf{Q}_{j-1}', \quad P_{\mathcal{S}_{j-1}^\perp} = \mathbf{I}_n - \mathbf{Q}_{j-1} \mathbf{Q}_{j-1}',$$

and the first identity is the Gram–Schmidt formula. Best approximation gives

$$\|\mathbf{v}_j - P_{\mathcal{S}_{j-1}} \mathbf{v}_j\| = \min_{\mathbf{s} \in \mathcal{S}_{j-1}} \|\mathbf{v}_j - \mathbf{s}\|.$$



Exercise 6.4. Let $\mathbf{A} = \mathbf{QR}$ be the thin QR factorisation of a full-column-rank matrix. Prove

$$\mathbf{A}'\mathbf{A} = \mathbf{R}'\mathbf{R}$$

and show that the normal equations for $\min_{\mathbf{x}} \|\mathbf{b} - \mathbf{A}\mathbf{x}\|^2$ are equivalent to

$$\mathbf{R}\mathbf{x} = \mathbf{Q}'\mathbf{b}.$$

Origin: H Ex. 14.23; H §12.4

Solution.

$$\mathbf{A}'\mathbf{A} = \mathbf{R}'\mathbf{Q}'\mathbf{Q}\mathbf{R} = \mathbf{R}'\mathbf{R}.$$

The normal equations reduce to

$$\mathbf{R}'\mathbf{R}\mathbf{x} = \mathbf{R}'\mathbf{Q}'\mathbf{b}.$$

The positive diagonal of $\mathbf{R}$ makes $\mathbf{R}'$ nonsingular, so multiplication by $(\mathbf{R}')^{-1}$ gives $\mathbf{R}\mathbf{x} = \mathbf{Q}'\mathbf{b}$; multiplication by $\mathbf{R}'$ proves the reverse implication. ■

Exercise 6.5. Let $\mathbf{P}$ be the orthogonal projector onto $\mathcal{S} \subseteq \mathbb{R}^n$ and define $\mathbf{R} = \mathbf{I}_n - 2\mathbf{P}$. Prove

$$\mathbf{R}' = \mathbf{R}, \quad \mathbf{R}'\mathbf{R} = \mathbf{I}_n, \quad \mathbf{R}\mathbf{s} = -\mathbf{s} \ (\mathbf{s} \in \mathcal{S}), \quad \mathbf{R}\mathbf{t} = \mathbf{t} \ (\mathbf{t} \in \mathcal{S}^\perp).$$

Origin: H Ex. 10.3

Solution. Since $\mathbf{P}' = \mathbf{P}^2 = \mathbf{P}$,

$$\mathbf{R}' = \mathbf{I}_n - 2\mathbf{P}' = \mathbf{R}, \quad \mathbf{R}'\mathbf{R} = \mathbf{R}^2 = \mathbf{I}_n - 4\mathbf{P} + 4\mathbf{P}^2 = \mathbf{I}_n.$$

For $\mathbf{s} \in \mathcal{S}$ and $\mathbf{t} \in \mathcal{S}^\perp$,

$$\mathbf{P}\mathbf{s} = \mathbf{s}, \quad \mathbf{P}\mathbf{t} = \mathbf{0},$$

which gives the two action identities. ■

Exercise 6.6. Let $\mathbf{E} \in \mathbb{R}^{n \times n}$ satisfy $\mathbf{E}^2 = \mathbf{E}$. Prove

$$\mathbb{R}^n = \text{col}(\mathbf{E}) \oplus \text{null}(\mathbf{E}).$$

Prove that $\mathbf{E}$ is symmetric if and only if $\text{col}(\mathbf{E}) \perp \text{null}(\mathbf{E})$.

Origin: S Ex. 3.5(a); H Thm. 12.3.9

Solution. For every $\mathbf{x}$,

$$\mathbf{x} = \mathbf{E}\mathbf{x} + (\mathbf{I}_n - \mathbf{E})\mathbf{x}, \quad \mathbf{E}\mathbf{x} \in \text{col}(\mathbf{E}), \quad \mathbf{E}(\mathbf{I}_n - \mathbf{E})\mathbf{x} = \mathbf{0}.$$

If $\mathbf{x} \in \text{col}(\mathbf{E}) \cap \text{null}(\mathbf{E})$, write $\mathbf{x} = \mathbf{E}\mathbf{a}$; then

$$\mathbf{x} = \mathbf{E}\mathbf{a} = \mathbf{E}^2\mathbf{a} = \mathbf{E}\mathbf{x} = \mathbf{0}.$$

This proves the direct sum. If $\mathbf{E}' = \mathbf{E}$, $\mathbf{s} = \mathbf{E}\mathbf{a}$, and $\mathbf{n} \in \text{null}(\mathbf{E})$, then

$$\mathbf{s}'\mathbf{n} = \mathbf{a}'\mathbf{E}'\mathbf{n} = \mathbf{a}'\mathbf{E}\mathbf{n} = 0.$$

Conversely, if the two summands are orthogonal, the displayed decomposition is the orthogonal decomposition onto $\text{col}(\mathbf{E})$ and $\mathbf{E}\mathbf{x}$ is its orthogonal projection. The projector characterisation gives $\mathbf{E}' = \mathbf{E}$. ■

Exercise 6.7. Let $\mathcal{T} \subseteq \mathcal{S} \subseteq \mathbb{R}^n$ with orthogonal projectors $\mathbf{P}_{\mathcal{T}}$ and $\mathbf{P}_{\mathcal{S}}$. Prove

$$\mathbf{P}_{\mathcal{T}}\mathbf{P}_{\mathcal{S}} = \mathbf{P}_{\mathcal{S}}\mathbf{P}_{\mathcal{T}} = \mathbf{P}_{\mathcal{T}}$$

and prove that $\mathbf{P}_{\mathcal{S}} - \mathbf{P}_{\mathcal{T}}$ is the orthogonal projector onto $\mathcal{S} \cap \mathcal{T}^{\perp}$. Deduce

$$\text{tr}(\mathbf{P}_{\mathcal{S}} - \mathbf{P}_{\mathcal{T}}) = \dim \mathcal{S} - \dim \mathcal{T}.$$

Origin: H Thm. 12.3.5; A&M Ex. 8.70

Solution. For every $\mathbf{x}$,

$$\mathbf{P}_{\mathcal{S}}\mathbf{P}_{\mathcal{T}}\mathbf{x} = \mathbf{P}_{\mathcal{T}}\mathbf{x}$$

because $\mathbf{P}_{\mathcal{T}}\mathbf{x} \in \mathcal{T} \subseteq \mathcal{S}$. Also $\mathbf{x} - \mathbf{P}_{\mathcal{S}}\mathbf{x} \in \mathcal{S}^{\perp} \subseteq \mathcal{T}^{\perp}$, so

$$\mathbf{P}_{\mathcal{T}}\mathbf{x} = \mathbf{P}_{\mathcal{T}}\mathbf{P}_{\mathcal{S}}\mathbf{x} + \mathbf{P}_{\mathcal{T}}(\mathbf{x} - \mathbf{P}_{\mathcal{S}}\mathbf{x}) = \mathbf{P}_{\mathcal{T}}\mathbf{P}_{\mathcal{S}}\mathbf{x}.$$

Put $\mathbf{D} = \mathbf{P}_{\mathcal{S}} - \mathbf{P}_{\mathcal{T}}$. Then

$$\mathbf{D}' = \mathbf{D}, \quad \mathbf{D}^2 = \mathbf{P}_{\mathcal{S}} - \mathbf{P}_{\mathcal{T}} - \mathbf{P}_{\mathcal{T}} + \mathbf{P}_{\mathcal{T}} = \mathbf{D}.$$

Thus $\mathbf{D}$ is an orthogonal projector. For every $\mathbf{x}$,

$$\mathbf{D}\mathbf{x} \in \mathcal{S}, \quad \mathbf{P}_{\mathcal{T}}\mathbf{D}\mathbf{x} = \mathbf{0},$$

so $\text{col}(\mathbf{D}) \subseteq \mathcal{S} \cap \mathcal{T}^{\perp}$. If $\mathbf{z} \in \mathcal{S} \cap \mathcal{T}^{\perp}$, then

$$\mathbf{D}\mathbf{z} = \mathbf{P}_{\mathcal{S}}\mathbf{z} - \mathbf{P}_{\mathcal{T}}\mathbf{z} = \mathbf{z},$$

which proves the reverse inclusion. Finally,

$$\operatorname{tr}(\mathbf{D}) = \operatorname{tr}(\mathbf{P}_{\mathcal{S}}) - \operatorname{tr}(\mathbf{P}_{\mathcal{T}}) = \dim \mathcal{S} - \dim \mathcal{T}.$$



Exercise 6.8. Let $\mathcal{S} \perp \mathcal{T}$, with orthogonal projectors $\mathbf{P}_{\mathcal{S}}$ and $\mathbf{P}_{\mathcal{T}}$. Prove that $\mathbf{P}_{\mathcal{S}} + \mathbf{P}_{\mathcal{T}}$ is the orthogonal projector onto $\mathcal{S} \oplus \mathcal{T}$, and deduce

$$\operatorname{rank}(\mathbf{P}_{\mathcal{S}} + \mathbf{P}_{\mathcal{T}}) = \operatorname{rank}(\mathbf{P}_{\mathcal{S}}) + \operatorname{rank}(\mathbf{P}_{\mathcal{T}}).$$

Origin: A&M Ex. 8.64

Solution. For $\mathbf{s} \in \mathcal{S}$, $\mathbf{t} \in \mathcal{T}$,

$$\mathbf{P}_{\mathcal{S}}\mathbf{t} = \mathbf{0}, \quad \mathbf{P}_{\mathcal{T}}\mathbf{s} = \mathbf{0},$$

so $\mathbf{P}_{\mathcal{S}}\mathbf{P}_{\mathcal{T}} = \mathbf{P}_{\mathcal{T}}\mathbf{P}_{\mathcal{S}} = \mathbf{0}$. Hence

$$(\mathbf{P}_{\mathcal{S}} + \mathbf{P}_{\mathcal{T}})' = \mathbf{P}_{\mathcal{S}} + \mathbf{P}_{\mathcal{T}}, \quad (\mathbf{P}_{\mathcal{S}} + \mathbf{P}_{\mathcal{T}})^2 = \mathbf{P}_{\mathcal{S}} + \mathbf{P}_{\mathcal{T}}.$$

Its column space is contained in $\mathcal{S} + \mathcal{T}$, and it fixes every $\mathbf{s} + \mathbf{t} \in \mathcal{S} + \mathcal{T}$. It is therefore the

orthogonal projector onto $\mathcal{S} \oplus \mathcal{T}$. The trace-rank identity gives

$$\text{rank}(\mathbf{P}_{\mathcal{S}} + \mathbf{P}_{\mathcal{T}}) = \text{tr}(\mathbf{P}_{\mathcal{S}} + \mathbf{P}_{\mathcal{T}}) = \text{rank}(\mathbf{P}_{\mathcal{S}}) + \text{rank}(\mathbf{P}_{\mathcal{T}}).$$



Exercise 6.9. Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ and $\mathbf{Y} \in \mathbb{R}^{n \times q}$. Prove that $\mathbf{B}_0 \in \mathbb{R}^{p \times q}$ minimises $\|\mathbf{Y} - \mathbf{XB}\|_F^2$ if and only if

$$\mathbf{X}'\mathbf{XB}_0 = \mathbf{X}'\mathbf{Y}.$$

Prove that $\mathbf{XB}_0 = \mathbf{HY}$ is unique even when $\mathbf{B}_0$ is not.

Origin: H Ex. 12.3

Solution. Write $\mathbf{Y} = (\mathbf{y}_1, \dots, \mathbf{y}_q)$ and $\mathbf{B} = (\boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_q)$. Then

$$\|\mathbf{Y} - \mathbf{XB}\|_F^2 = \sum_{j=1}^q \|\mathbf{y}_j - \mathbf{X}\boldsymbol{\beta}_j\|^2.$$

The sum is minimised if and only if every summand is minimised. The normal-equation theorem

gives

$$(\forall j) \mathbf{X}'\mathbf{X}\boldsymbol{\beta}_j = \mathbf{X}'\mathbf{y}_j \iff \mathbf{X}'\mathbf{X}\mathbf{B} = \mathbf{X}'\mathbf{Y}.$$

For every minimiser and every j ,

$$\mathbf{X}\boldsymbol{\beta}_j = \mathbf{H}\mathbf{y}_j,$$

so $\mathbf{X}\mathbf{B} = \mathbf{H}\mathbf{Y}$ is unique. 

Exercise 6.10. Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ and $\mathbf{Z} \in \mathbb{R}^{n \times q}$ satisfy $\text{col}(\mathbf{X}) = \text{col}(\mathbf{Z})$. Prove that, for every response $\mathbf{y}$, least squares with $\mathbf{X}$ and least squares with $\mathbf{Z}$ have the same fitted vector, residual vector, and minimum residual sum of squares. If

$$\mathbf{X} = \mathbf{ZF}, \quad \mathbf{Z} = \mathbf{XG},$$

and β_0, γ_0 are respective minimisers, prove that the two coefficient minimiser sets are

$$\beta_0 + \text{null}(\mathbf{X}), \quad \gamma_0 + \text{null}(\mathbf{Z}).$$

Prove also that $\mathbf{F}$ maps every $\mathbf{X}$ -coefficient minimiser to a $\mathbf{Z}$ -coefficient minimiser, while $\mathbf{G}$ maps in the reverse direction.

Origin: H Thm. 12.4.3 and Lemma 4.2.2

Solution. The common column space has one orthogonal projector $\mathbf{H}$. In both problems,

$$\hat{\mathbf{y}} = \mathbf{Hy}, \quad \hat{\boldsymbol{\varepsilon}} = (\mathbf{I}_n - \mathbf{H})\mathbf{y},$$

Moreover,

$$\min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \|(\mathbf{I}_n - \mathbf{H})\mathbf{y}\|^2 = \min_{\boldsymbol{\gamma}} \|\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma}\|^2.$$

The normal-equation theorem gives the coefficient minimiser sets

$$\boldsymbol{\beta}_0 + \text{null}(\mathbf{X}), \quad \boldsymbol{\gamma}_0 + \text{null}(\mathbf{Z}).$$

If $\hat{\boldsymbol{\beta}}$ minimises the $\mathbf{X}$ problem, then

$$\mathbf{Z}(\mathbf{F}\hat{\boldsymbol{\beta}}) = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{H}\mathbf{y},$$

so $\mathbf{F}\hat{\boldsymbol{\beta}}$ minimises the $\mathbf{Z}$ problem. If $\hat{\boldsymbol{\gamma}}$ minimises the $\mathbf{Z}$ problem, then

$$\mathbf{X}(\mathbf{G}\hat{\boldsymbol{\gamma}}) = \mathbf{Z}\hat{\boldsymbol{\gamma}} = \mathbf{H}\mathbf{y},$$

so $\mathbf{G}\hat{\boldsymbol{\gamma}}$ minimises the $\mathbf{X}$ problem. 

Exercise 6.11. Let $\mathcal{T} \subseteq \mathcal{S} \subseteq \mathbb{R}^n$ and define

$$\text{RSS}_{\mathcal{S}}(\mathbf{y}) = \|(\mathbf{I}_n - \mathbf{P}_{\mathcal{S}})\mathbf{y}\|^2, \quad \text{RSS}_{\mathcal{T}}(\mathbf{y}) = \|(\mathbf{I}_n - \mathbf{P}_{\mathcal{T}})\mathbf{y}\|^2.$$

Prove

$$\text{RSS}_{\mathcal{T}}(\mathbf{y}) - \text{RSS}_{\mathcal{S}}(\mathbf{y}) = \|(\mathbf{P}_{\mathcal{S}} - \mathbf{P}_{\mathcal{T}})\mathbf{y}\|^2 \geq 0,$$

and determine when equality holds.

Origin: CLM Ex. 11.14(a); H Thm. 12.4.1

Solution. The three vectors

$$\mathbf{P}_{\mathcal{T}}\mathbf{y}, \quad (\mathbf{P}_{\mathcal{S}} - \mathbf{P}_{\mathcal{T}})\mathbf{y}, \quad (\mathbf{I}_n - \mathbf{P}_{\mathcal{S}})\mathbf{y}$$

belong respectively to $\mathcal{T}$, $\mathcal{S} \cap \mathcal{T}^\perp$, and $\mathcal{S}^\perp$, so they are pairwise orthogonal. Since

$$(\mathbf{I}_n - \mathbf{P}_{\mathcal{T}})\mathbf{y} = (\mathbf{P}_{\mathcal{S}} - \mathbf{P}_{\mathcal{T}})\mathbf{y} + (\mathbf{I}_n - \mathbf{P}_{\mathcal{S}})\mathbf{y},$$

Pythagoras gives the identity. Equality holds exactly when

$$(\mathbf{P}_{\mathcal{S}} - \mathbf{P}_{\mathcal{T}})\mathbf{y} = \mathbf{0} \iff \mathbf{P}_{\mathcal{S}}\mathbf{y} = \mathbf{P}_{\mathcal{T}}\mathbf{y}. \quad \blacksquare$$

Exercise 6.12. Let $(\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3)$ have full column rank and put $\mathbf{C} = (\mathbf{X}_2, \mathbf{X}_3)$. Let $\mathbf{P}_3$ project onto $\text{col}(\mathbf{X}_3)$, put $\mathbf{M}_3 = \mathbf{I}_n - \mathbf{P}_3$, and let $\mathbf{P}_{23}$ project onto $\text{col}(\mathbf{C})$. If $\mathbf{W} = \mathbf{M}_3\mathbf{X}_2$, let $\mathbf{P}_\mathbf{W}$ project onto $\text{col}(\mathbf{W})$, and put $\mathbf{M}_\mathbf{W} = \mathbf{I}_n - \mathbf{P}_\mathbf{W}$. Prove

$$\mathbf{I}_n - \mathbf{P}_{23} = \mathbf{M}_\mathbf{W}\mathbf{M}_3 = \mathbf{M}_3\mathbf{M}_\mathbf{W}.$$

Deduce that the coefficient on $\mathbf{X}_1$ can be obtained after successively residualising $\mathbf{X}_1$ and $\mathbf{y}$ on $\mathbf{X}_3$ and then on $\mathbf{W}$.

Origin: CLM Thm. 8.1 and Ex. 11.18

Solution. Apply the FWL decomposition to $(\mathbf{X}_2, \mathbf{X}_3)$:

$$\mathbf{P}_{23} = \mathbf{P}_3 + \mathbf{P}_\mathbf{W}, \quad \mathbf{P}_3\mathbf{P}_\mathbf{W} = \mathbf{P}_\mathbf{W}\mathbf{P}_3 = \mathbf{O}.$$

Therefore

$$\mathbf{I}_n - \mathbf{P}_{23} = (\mathbf{I}_n - \mathbf{P}_\mathbf{W})(\mathbf{I}_n - \mathbf{P}_3) = (\mathbf{I}_n - \mathbf{P}_3)(\mathbf{I}_n - \mathbf{P}_\mathbf{W}).$$

Put $\mathbf{M}_{23} = \mathbf{I}_n - \mathbf{P}_{23}$, $\mathbf{X}_1^* = \mathbf{M}_{23}\mathbf{X}_1$, and $\mathbf{y}^* = \mathbf{M}_{23}\mathbf{y}$. FWL applied to $(\mathbf{X}_1, \mathbf{C})$ gives

$$(\mathbf{X}_1^*)'\mathbf{X}_1^* = \mathbf{X}_1'\mathbf{M}_{23}\mathbf{X}_1, \quad (\mathbf{X}_1^*)'\mathbf{y}^* = \mathbf{X}_1'\mathbf{M}_{23}\mathbf{y},$$

$$\widehat{\boldsymbol{\beta}}_1 = (\mathbf{X}'_1 \mathbf{M}_{23} \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{M}_{23} \mathbf{y} = ((\mathbf{X}_1^*)' \mathbf{X}_1^*)^{-1} (\mathbf{X}_1^*)' \mathbf{y}^*.$$

The identity $\mathbf{M}_{23} = \mathbf{M}_w \mathbf{M}_3$ is the asserted successive residualisation. ■

Exercise 6.13. Let $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)$ have full column rank and suppose $\mathbf{X}_1$ has full column rank. Let $\widetilde{\boldsymbol{\beta}}_1$ be the coefficient from least squares of $\mathbf{y}$ on $\mathbf{X}_1$ alone, and let $(\widehat{\boldsymbol{\beta}}'_1, \widehat{\boldsymbol{\beta}}'_2)'$ be the coefficient from least squares on $\mathbf{X}$. Prove the deterministic identity

$$\widetilde{\boldsymbol{\beta}}_1 = \widehat{\boldsymbol{\beta}}_1 + (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{X}_2 \widehat{\boldsymbol{\beta}}_2.$$

Origin: CLM Thm. 8.2 and Ex. 11.19

Solution. The full residual

$$\widehat{\boldsymbol{\varepsilon}} = \mathbf{y} - \mathbf{X}_1 \widehat{\boldsymbol{\beta}}_1 - \mathbf{X}_2 \widehat{\boldsymbol{\beta}}_2$$

satisfies $\mathbf{X}'_1 \widehat{\boldsymbol{\varepsilon}} = \mathbf{0}$. Hence

$$\begin{aligned}\widetilde{\boldsymbol{\beta}}_1 &= (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{y} \\ &= (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 (\mathbf{X}_1 \widehat{\boldsymbol{\beta}}_1 + \mathbf{X}_2 \widehat{\boldsymbol{\beta}}_2 + \widehat{\boldsymbol{\varepsilon}}) \\ &= \widehat{\boldsymbol{\beta}}_1 + (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{X}_2 \widehat{\boldsymbol{\beta}}_2.\end{aligned}$$

Exercise 6.14. Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ have rank r and satisfy $\mathbf{1} \in \text{col}(\mathbf{X})$. Prove

$$\frac{1}{n} \leq h_{ii} \leq 1, \quad i = 1, \dots, n.$$

Also prove

$$\min_i h_{ii} \leq \frac{r}{n} \leq \max_i h_{ii}.$$

Origin: CLM Thm. 2.5 and Ex. 11.12(a)–(b)

Solution. The unit vector $\mathbf{q}_1 = n^{-1/2}\mathbf{1}$ extends to an orthonormal basis $\mathbf{q}_1, \dots, \mathbf{q}_r$ of $\text{col}(\mathbf{X})$. Therefore

$$h_{ii} = \sum_{j=1}^r q_{ij}^2 = \frac{1}{n} + \sum_{j=2}^r q_{ij}^2 \geq \frac{1}{n}.$$

The upper bound was proved in the leverage theorem. Since

$$\frac{1}{n} \sum_{i=1}^n h_{ii} = \frac{r}{n},$$

the mean lies between the minimum and maximum. 

Exercise 6.15. Let $(\mathbf{1}, \mathbf{Z}) \in \mathbb{R}^{n \times (q+1)}$ have full column rank. Put

$$\mathbf{M}_1 = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}', \quad \bar{y} = \frac{1}{n}\mathbf{1}'\mathbf{y}, \quad \bar{\mathbf{z}}' = \frac{1}{n}\mathbf{1}'\mathbf{Z}.$$

For least squares of $\mathbf{y}$ on $(\mathbf{1}, \mathbf{Z})$, prove

$$\hat{\boldsymbol{\beta}}_Z = (\mathbf{Z}'\mathbf{M}_1\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{M}_1\mathbf{y}, \quad \hat{\beta}_0 = \bar{y} - \bar{\mathbf{z}}'\hat{\boldsymbol{\beta}}_Z, \quad \mathbf{1}'\hat{\boldsymbol{\varepsilon}} = 0.$$

Origin: A&M Ex. 8.71(a)–(c); CLM Thm. 8.1

Solution. The projector onto $\text{span}\{\mathbf{1}\}$ is $n^{-1}\mathbf{1}\mathbf{1}'$, so FWL with $\mathbf{X}_1 = \mathbf{Z}$ and $\mathbf{X}_2 = \mathbf{1}$ gives

$$\hat{\boldsymbol{\beta}}_Z = (\mathbf{Z}'\mathbf{M}_1\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{M}_1\mathbf{y}.$$

The intercept normal equation is

$$0 = \mathbf{1}'(\mathbf{y} - \mathbf{1}\hat{\beta}_0 - \mathbf{Z}\hat{\boldsymbol{\beta}}_Z) = n\bar{y} - n\hat{\beta}_0 - n\bar{\mathbf{z}}'\hat{\boldsymbol{\beta}}_Z,$$

which gives the intercept formula and $\mathbf{1}'\hat{\boldsymbol{\varepsilon}} = 0$. 